

MIDTERM PROJECT PROPOSAL

PaperReady - An Agentic RAG System for Pre-Submission Manuscript Validation

INTRODUCTION

METHODOLOGY

EXPECTED RESULT

SUBJECT : Big Data Analysis (8862016-01)

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01

Problem Definition & Motivation

Problem and Goal

Problem Definition

- **Researchers** and **graduate students** preparing a paper for **journal submission** must follow many journal-specific rules that found in long author guideline PDFs.
- These include **title page format**, **ORCID**, **reference style**, **figure captions**, **cover letter format**, **conflict of interest statements**, and **more**.
- The rules are different for each journal. Manual checking is **slow**, **easy to get wrong**, and can **lead to desk rejection**. Authors spend many hours checking documents but may still miss important requirements.

Motivation

- Build an **Agentic RAG system** that helps check a paper before journal submission. The agent reads the target **journal author guidelines PDF**, the **user manuscript draft**, and a **structured database** of journal information and **reference style rules**.
- The Agentic RAG system can:
 - Check the manuscript section by section.
 - Create a checklist of missing items.
 - Verify references using external APIs.
 - Draft a cover letter based only on the journal's real requirements.
- The goal is to build **practical problem-solving** skills by combining **LLM reasoning**, **RAG**, **structured data** lookup, and **external tool calls**.

02

System Architecture

Agent Brain, Orchestration and Workflow

System Architecture

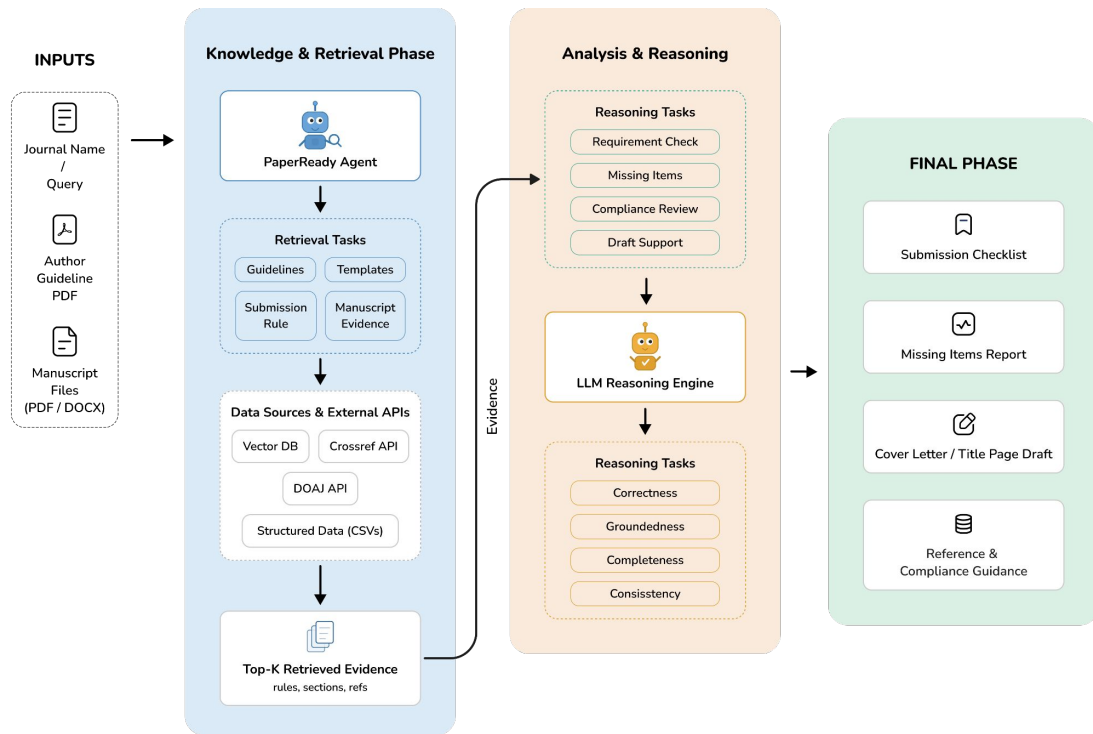


Figure 1. System Architecture: RAG-Based Academic Submission Assistant Agent.

03

Data Source

Structured Data, Documents and Web-Based

Data Sources - Structured Data

- We have two **tables**:
 - **Table 1 - journal_metadata.csv**: contains **50+ academic journals** with fields such as
 - Name
 - Publisher
 - Scope keywords
 - Impact factor
 - Review time
 - Submission fee
 - Open-access status
 - Required reference style.
 - Data is sourced from **SCImago Journal Rank** [\[1\]](#) and **DOAJ** [\[2\]](#) (both free and public).

Data Sources - Structured Data

- We have two **tables**:
 - **Table 2 - reference_style_rules.csv**: contains **50+ entries** defining formatting rules for major reference styles (APA 7, IEEE, Vancouver, Harvard, Chicago). Fields include:
 - style name
 - element type (author, year, title, journal)
 - Order
 - Italics or capitalization.
 - The data is compiled from **official style guides**.

Data Sources - Documents & External Tools

- We have two **documents**:
 - **Doc 1 - Journal Author Guidelines PDF**: for example, *Image and Vision Computing*: Guide for Authors (~12 pages), freely downloadable from Elsevier [\[3\]](#).
 - **Doc 2 - Manuscript Template / Sample Article PDF**: a representative published article from the same journal (~10 pages), used as a structural reference for a compliant submission [\[4\]](#).
- We have two **external API sources**:
 - **Source 1 - Crossref REST API**: verifies that all DOIs in the manuscript references link to real publications. Free, no API key required [\[5\]](#).
 - **Source 2 - DOAJ API**: confirms whether a journal is legitimate, peer-reviewed, and open-access, and checks article processing charges. Free, no API key required [\[6\]](#).

04

Team Member Responsibilities

Members Role

Task Handling

- **CHUUN Rotanakkosal:**
 - Integrate the **LLM** using **Gemini 2.5 Pro/ Flash** through the Google AI Studio API.
 - Develop the **RAG pipeline** by chunking and embedding the **journal guideline PDF** and **template document**, then indexing them in **ChromaDB**.
 - Design system prompt for the agent reasoning stages: **validation, checklist generation, ...** etc.
- **VORN Naro:**
 - Integrate the **external APIs** (Crossref and DOAJ) and implement the **agent function-calling logic** to select the correct tool for query
 - Build the **structured-data look up** layer for **journal_metadata.csv** and **reference_style.csv** using pandas.
 - Build demo UI using **streamlit**.
 - Create question for **Q&A evaluation pair** with gold-standard to produce the result table.

05

References

Accessible References Used in the Presentation

1. SCImago Journal & Country Rank [\[1\]](#).
2. DOAJ Open Global Trusted [\[2\]](#).
3. Journal Author Guidelines PDF [\[3\]](#).
4. Sample Article PDF [\[4\]](#).
5. Crossref REST AP [\[5\]](#).
6. DOAJ API [\[6\]](#).



Thank You!

